Holomorphy of the Dirichlet-to-Neumann semigroup on continuous boundary functions for Lipschitz domains

Kaiwen Jin*

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Abstract

Let $\Omega \subset \mathbb{R}^d$, $d \geq 2$, be a bounded connected Lipschitz domain, and let C be a real symmetric uniformly elliptic matrix with Lipschitz entries on $\overline{\Omega}$. We prove that the weak Dirichlet-to-Neumann semigroup on $C(\partial\Omega)$ is holomorphic of angle $\pi/2$ and bounded on every strictly smaller sector. This answers the continuous-function question in the real symmetric case of Question 4 of ter Elst and Ouhabaz. The proof obtains a supremum norm resolvent bound by boundary rescaling. A complex Robin uniqueness lemma on a half-space excludes the limiting solution; its proof uses vertical L^2 Dirichlet estimates and an iteration of boundary mass bounds.

1 Introduction

The Dirichlet-to-Neumann operator describes diffusion on the boundary through harmonic extension into the domain. Its realization on L^2 is nonnegative and self-adjoint when the underlying elliptic coefficients are real and symmetric. Holomorphy on continuous boundary functions requires an estimate in the supremum norm, which does not follow from this Hilbert space fact.

Throughout, $\Omega \subset \mathbb{R}^d$, $d \geq 2$, is a bounded connected Lipschitz domain, $\Gamma = \partial\Omega$, and σ is $(d-1)$-dimensional Hausdorff measure on Γ. The matrix $C = (c_{kl})$ is real symmetric, its entries are Lipschitz on $\overline{\Omega}$, and

$$\kappa|\xi|^2 \leq \xi \cdot C(x)\xi \leq K|\xi|^2 \qquad (x \in \overline{\Omega},\ \xi \in \mathbb{R}^d) \tag{1}$$

for some $0 < \kappa \leq K < \infty$. All function spaces below are complex. The weak Dirichlet-to-Neumann operator $\mathcal{N}$ on $L^2(\Gamma, \sigma)$ is defined by $\varphi \in D(\mathcal{N})$ and $\mathcal{N}\varphi = h$ if there is $u \in H^1(\Omega)$ with $\operatorname{Tr} u = \varphi$ such that

$$\int_\Omega C\nabla u \cdot \nabla \overline{v}\,\mathrm{d}x = \int_\Gamma h\,\overline{\operatorname{Tr} v}\,\mathrm{d}\sigma \qquad (v \in H^1(\Omega)). \tag{2}$$

In particular, u is harmonic for $-\operatorname{div}(C\nabla)$, and is the unique harmonic extension of φ. The sign in (2) is that of the outward conormal derivative. Write $\mathcal{N}_c$ for the part of $\mathcal{N}$ in $C(\Gamma)$:

$$D(\mathcal{N}_c) = \{\varphi \in D(\mathcal{N}) \cap C(\Gamma) : \mathcal{N}\varphi \in C(\Gamma)\}.$$

We equip $C(\Gamma)$ with its usual supremum norm.

Arendt and ter Elst [1] proved that $-\mathcal{N}_c$ generates a positive contractive C_0-semigroup T and that $T(t)$ has a continuous kernel for $t > 0$. Their result identifies this semigroup

*Contact: yui.ui.siki@gmail.com.

with the restriction of the L^2 semigroup. On $C^{1+\alpha}$ domains, ter Elst and Ouhabaz [3] established holomorphy of angle $\pi/2$ under Hölder regularity of real symmetric coefficients. The final paragraph of Question 4 in [4] asks about holomorphy on $C(\Gamma)$ when the boundary is only Lipschitz and the real coefficients are Lipschitz. We prove the following result for the symmetric class above.

For $0 < \theta \le \pi/2$, put $\Sigma_\theta = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \theta\}$.

Theorem 1.1. *Under (1) and the preceding hypotheses, T extends to an operator norm holomorphic semigroup on $\Sigma_{\pi/2}$. For every $0 < \theta < \pi/2$ there is a finite constant M_θ, depending on Ω, C, θ, such that*

$$\sup_{z \in \Sigma_\theta} \|T(z)\|_{C(\Gamma)\to C(\Gamma)} \le M_\theta. \tag{3}$$

For every $\varphi \in C(\Gamma)$, $T(z)\varphi \to \varphi$ as $z \to 0$ in any strictly smaller sector. No connectedness assumption on Γ is needed.

For every $\varepsilon \in (0, \pi)$ there is a finite M_ε such that

$$\|\lambda(\lambda + \mathcal{N}_c)^{-1}\|_{C(\Gamma)\to C(\Gamma)} \le M_\varepsilon \qquad (\lambda \ne 0,\ |\arg \lambda| \le \pi - \varepsilon). \tag{4}$$

If this estimate fails at large $|\lambda|$, rescaling at a point of maximum boundary value gives a bounded solution of a complex Robin problem on an unbounded Lipschitz graph. The original coefficient field becomes constant under rescaling. Flattening the limiting graph then gives a uniformly elliptic matrix independent of the normal variable. For that matrix, the half-space L^2 Dirichlet theorem of Auscher, Axelsson and McIntosh [2] controls the mass of a bounded harmonic function above a large boundary ball. Combining this estimate with the Robin energy identity repeatedly improves the power of the boundary mass until the limiting solution vanishes.

The graph area densities need not converge to the area density of the limiting graph. We keep their weak-$*$ limit as a separate positive bounded weight in the Robin condition. This is why the half-space uniqueness result below includes such a weight.

2 Half-space estimates and complex Robin uniqueness

Let $n \ge 1$, $\mathbb{H} = \mathbb{R}^n \times (0, \infty)$, and write its coordinates as (x, t). In this section $A = A(x)$ is a bounded measurable real symmetric $(n+1) \times (n+1)$ matrix, independent of t, satisfying uniform ellipticity as in (1). We use $B'(q, R)$ for a ball in $\mathbb{R}^n$ and

$$Q(q, R) = B'(q, R) \times (0, R).$$

The phrase $H^1_{\mathrm{loc}}(\overline{\mathbb{H}})$ means H^1 on every bounded half-space patch, including its flat boundary. Compactly supported tests in this space are allowed to have a nonzero boundary trace.

2.1 Dirichlet solutions of finite energy

Let $\dot{H}^1_0(\mathbb{H})$ be the completion of $C_c^\infty(\mathbb{H})$ for the gradient norm. Hardy's inequality

$$\|W/t\|_{L^2(\mathbb{H})} \le 2\|\partial_t W\|_{L^2(\mathbb{H})} \tag{5}$$

identifies its elements as local Sobolev functions and fixes their additive constant. A local Sobolev function with zero trace and finite gradient norm belongs to this space. Indeed, vertical integration by parts on $(0, R)$, with a nonpositive upper boundary term, proves (5);

radial cutoffs and then cutoffs near $t = 0$ have errors controlled by the tails of $|\nabla W|^2 + |W/t|^2$. Mollification of the remaining compact interior functions gives the assertion.

For compactly supported $f \in H^{1/2}(\mathbb{R}^n)$, its usual Poisson extension $P_t f$ has finite gradient norm. With the unitary Fourier transform,

$$\int_{\mathbb{H}} |\nabla P_t f|^2 \, \mathrm{d}x \, \mathrm{d}t = \int_{\mathbb{R}^n} |\xi| |\widehat{f}(\xi)|^2 \, \mathrm{d}\xi. \tag{6}$$

This follows by integrating the multiplier $e^{-t|\xi|}$ in t and using Plancherel [7, Chapter 9]. The Poisson kernel is nonnegative, has total mass one, and contracts L^2 and L^∞. Lax–Milgram on $\dot{H}^1_0(\mathbb{H})$ supplies a unique A-harmonic function $H_A f$ such that $H_A f - P_t f \in \dot{H}^1_0(\mathbb{H})$. We use this notation only for the variational solution just defined.

Lemma 2.1. *There is a finite constant K_A such that, for every compactly supported $f \in H^{1/2}(\mathbb{R}^n)$,*

$$\sup_{t>0} \|H_A f(\cdot, t)\|_{L^2(\mathbb{R}^n)} \le K_A \|f\|_{L^2(\mathbb{R}^n)}. \tag{7}$$

If also $f \in L^\infty$, then

$$\|H_A f\|_{L^\infty(\mathbb{H})} \le \|f\|_{L^\infty(\mathbb{R}^n)}. \tag{8}$$

Proof. For smooth compactly supported f, the solution agrees with the L^2 Dirichlet solution of [2], by Theorem 5.3 there. Its connected-component hypothesis holds because $(1-s)I + sA$, $0 \le s \le 1$, stays in the real symmetric uniformly elliptic class; this class is well posed by their Theorem 2.2(i). Their Corollary 2.4 gives (7). The homogeneous zero-trace space in their Definition 5.1 agrees with the completion above by (5) and the cutoff argument. Their ordering (t, x) is changed to (x, t) by an orthogonal permutation.

Approximate general f by smooth compactly supported f_k in $H^{1/2}$. Formula (6) and the variational equation give convergence of the gradients. Hardy's inequality for the zero-trace differences then gives local H^1 convergence up to $t = 0$. Applied to differences, the L^2 Dirichlet bound also gives convergence in $\sup_t L^2$ norm. These limits agree locally, proving (7) for the variational solution.

For the maximum principle, first take real f and set $M = \|f\|_\infty$. If $U = H_A f$, then $w = (U - M)^+$ has finite gradient norm, zero trace, and $w \le |U - P_t f|$, since $P_t f \le M$. Thus $w/t \in L^2$, and the same cutoff argument shows $w \in \dot{H}^1_0$. Testing the equation with w gives $\int A \nabla w \cdot \nabla w = 0$, so $w = 0$. The lower bound follows in the same way. For complex data apply the upper bound to $\mathrm{Re}(e^{-i\alpha} U)$ for a countable dense set of phases α. This yields (8). □

We will also need uniqueness without a global energy assumption.

Lemma 2.2. *A bounded A-harmonic function in $H^1_{\mathrm{loc}}(\overline{\mathbb{H}})$ with zero trace is zero. The same assertion holds when the real uniformly elliptic coefficients depend on both x and t.*

Proof. Reflect the function oddly across $t = 0$ and the matrix by $A_-(x, t) = JA(x, -t)J$, where $J = \mathrm{diag}(I_n, -1)$. For a whole-space test η, the upper-half-space test $\eta(x, t) - \eta(x, -t)$ has zero trace. It proves the weak equation for the odd extension, whose matrix remains real uniformly elliptic. The interior Hölder estimate for such equations [9, Corollary 5.5] gives, for each fixed ball B_r and arbitrarily large R, an oscillation bound $C(r/R)^\alpha \|U\|_\infty$. Hence each real component of the extension is constant. Oddness makes the constant zero. □

2.2 From boundary mass to bulk mass

Lemma 2.3. *Let $U \in H^1_{\mathrm{loc}}(\overline{\mathbb{H}})$ be bounded and A-harmonic, and write $\varphi = \operatorname{Tr} U$. Suppose that, for some $\beta < n$ and finite M_β,*

$$F(R) := \sup_{q\in\mathbb{R}^n} \int_{B'(q,R)} |\varphi|^2 \,\mathrm{d}x \le M_\beta R^\beta \qquad (R \ge 1). \tag{9}$$

Then there is a finite constant C_β such that

$$\sup_{q\in\mathbb{R}^n} \int_{Q(q,R)} |U|^2 \,\mathrm{d}x\,\mathrm{d}t \le C_\beta R^{\beta+1} \qquad (R \ge 1). \tag{10}$$

Proof. Fix q, R and a nonincreasing radial Lipschitz cutoff χ, equal to one on B'_1 and zero outside B'_2. Set

$$\chi_0(x) = \chi\left(\frac{x-q}{4R}\right), \qquad \chi_j(x) = \chi\left(\frac{x-q}{2^j 4R}\right) - \chi\left(\frac{x-q}{2^{j-1}4R}\right) \quad (j \ge 1).$$

The χ_j are nonnegative and sum to one. The far piece χ_j vanishes on $B'(q, S_j)$, where $S_j = 2^{j-1}4R$. Put $f_j = \chi_j\varphi$ and $U_j = H_A f_j$. Every f_j is compactly supported $H^{1/2}$ data: apply the local H^1 trace theorem to a tangential and vertical cutoff of U.

For the near piece, Lemma 2.1 gives

$$\int_{Q(q,R)} |U_0|^2 \le RK_A^2 \|f_0\|_2^2 \le CR^{\beta+1}. \tag{11}$$

For $j \ge 1$, odd reflection of U_j gives a homogeneous elliptic equation in the ball of radius S_j centered at $(q, 0)$. The interior L^2-to-supremum estimate [9, Theorem 5.1], together with (7), implies

$$\begin{aligned} \sup_{Q(q,R)} |U_j| &\le C S_j^{-(n+1)/2} \|U_j\|_{L^2(B'(q,S_j)\times(0,S_j))} \\ &\le C S_j^{-n/2} \|f_j\|_2 \le C_\beta S_j^{(\beta-n)/2}. \end{aligned} \tag{12}$$

Here $Q(q, R)$ lies inside the half-radius ball because $S_j \ge 4R$. Since $\beta < n$, the geometric series in (12) sums to at most $C_\beta R^{(\beta-n)/2}$.

The sum $V = \sum_{j\ge0} U_j$ converges uniformly on compact patches for its far tail. It also converges in local H^1 up to the boundary. To see the latter assertion, fix a larger patch; sufficiently far finite tails have zero trace there. Testing their equation with the tail times a squared cutoff gives the boundary Caccioppoli estimate

$$\|\nabla(\text{tail})\|^2_{L^2(\text{smaller patch})} \le C\|\text{tail}\|^2_{L^2(\text{larger patch})}.$$

The uniform tail bound makes the right-hand side tend to zero. Thus V is harmonic with trace φ. By linearity its partial sums equal $H_A((\sum_{j\le k}\chi_j)\varphi)$, and (8) bounds every partial sum by $\|\varphi\|_\infty$. Therefore V is bounded and Lemma 2.2 identifies it with U. Combining (11) with the square of the summed far bound and the volume $O(R^{n+1})$ proves (10). □

2.3 A Robin Liouville lemma

Proposition 2.4. *Let $b \in L^\infty(\mathbb{R}^n)$ satisfy $0 < b_* \le b \le b^* < \infty$. If $\lambda \in \mathbb{C} \setminus (-\infty, 0]$ and a bounded $U \in H^1_{\mathrm{loc}}(\overline{\mathbb{H}})$ satisfies*

$$\int_{\mathbb{H}} A\nabla U \cdot \nabla\overline{V} \,\mathrm{d}x\,\mathrm{d}t + \lambda \int_{\mathbb{R}^n} b\,\varphi\,\overline{\operatorname{Tr} V}\,\mathrm{d}x = 0, \qquad \varphi = \operatorname{Tr} U, \tag{13}$$

for all compactly supported local H^1 tests, then $U = 0$.

Proof. Write $\alpha = \arg\lambda \in (-\pi, \pi)$. Choose a cutoff η_R equal to one on $B'(q,R) \times [0,R]$, supported in $B'(q,2R) \times [0,2R]$, with $|\nabla \eta_R| \le C/R$. Use $V = \eta_R^2 U$ in (13), multiply by $e^{-i\alpha/2}$, and take real parts. Both the gradient quadratic form and the boundary term are coercive, with the common positive factor $\cos(\alpha/2)$. Young's inequality absorbs the cross term, giving

$$\int_{B'(q,R)} |\varphi|^2 \, \mathrm{d}x \le C_\lambda R^{-2} \int_{Q(q,2R)} |U|^2 \, \mathrm{d}x \, \mathrm{d}t. \tag{14}$$

The constants are finite for the fixed A, b, λ.

Boundedness of U first gives $F(R) \le CR^{n-1}$ for $R \ge 1$. This exponent is strictly below n. Whenever $F(R) \le C_\beta R^\beta$ with $\beta < n$, Lemma 2.3 and (14) improve it to $F(R) \le C_{\beta-1} R^{\beta-1}$. After n improvements we obtain $F(R) \le C/R$. For any fixed ball, its trace mass is bounded by $F(R)$ for all sufficiently large R, so it is zero. Hence $\varphi = 0$, and Lemma 2.2 gives $U = 0$. □

Remark 2.5. *The parameter exclusions in Proposition 2.4 are necessary. At $\lambda = 0$ constants are allowed. If $A = I$, $b = 1$, and $\lambda = -|\xi| < 0$, then $U(x,t) = e^{i\xi\cdot x} e^{-|\xi|t}$ is a nonzero bounded solution.*

3 Boundary compactness and the resolvent estimate

3.1 Local Neumann estimates on Lipschitz graphs

The interior estimates used above are the uniformly elliptic specializations of [9, Theorem 5.1 and Corollaries 5.4–5.5]. For the inhomogeneous equation $\operatorname{div}(B\nabla w) = \operatorname{div} F$, bounded F gives local Hölder continuity. After rescaling to fixed balls, the constants depend only on dimension, ellipticity and the forcing bound. The results apply separately to the real and imaginary parts.

Lemma 3.1. *Consider a Lipschitz graph $D = \{(x,s) : s > \psi(x)\}$ with $\operatorname{Lip}\psi \le L$, a real uniformly elliptic matrix field B_0 with fixed bounds, and a harmonic $v \in H^1_{\mathrm{loc}}(\overline{D})$ satisfying $\|v\|_\infty \le 1$. Suppose that its weak outward conormal derivative h is bounded by H on the graph patch in question. On each smaller patch, v has local H^1 and Hölder bounds depending only on d, L, the ellipticity bounds, H and the patch radii. It has ambient extensions with the same local bounds across the graph.*

Proof. Test the weak equation with $\eta^2 v$. The boundary term is bounded by H times the surface area of the patch, and Young's inequality absorbs the gradient cross term. This proves the local energy bound.

Flatten by $\Phi(x,t) = (x, t + \psi(x))$. The Sobolev chain rule and graph reflection are valid for Lipschitz ψ; see [6, arXiv version, Section 3.1 and Lemma 3.3]. The determinant of $D\Phi$ is one, and the transformed matrix is

$$B = (D\Phi)^{-1} B_0(\Phi) (D\Phi)^{-T}.$$

It is real uniformly elliptic with bounds controlled by L. The graph area formula [5, Section 3.3.4] gives $a(x) = \sqrt{1 + |\nabla\psi(x)|^2}$. Writing $\widetilde{h}(x) = h(x,\psi(x))$, the bounded field $F = (0, \dots, 0, -\widetilde{h}a)$, independent of t, satisfies

$$\int_{\mathbb{H}} F \cdot \nabla \overline{V} = \int_{\mathbb{R}^{d-1}} \widetilde{h} a \, \overline{\operatorname{Tr} V}.$$

Thus the flattened weak equation is $\int B\nabla(v\circ\Phi)\cdot\nabla\overline{V} = \int F\cdot\nabla\overline{V}$. Extend $v\circ\Phi$ evenly across $t = 0$, the matrix by JBJ, and the field by JF, where $J = \operatorname{diag}(I_{d-1}, -1)$. Testing

with the sum of the reflected tests proves the whole-space inhomogeneous equation. The interior Hölder estimate now applies. Transforming back and using the same reflection gives the claimed ambient H^1 and Hölder extensions. □

3.2 Compact resolvents and small parameters

We first recall why all the resolvents used in (4) exist and why only large parameters can violate that estimate. The continuous kernel theorem and the C_0 theorem in [1, Corollary 3.5 and Theorem 5.3(a)] apply with potential $V = 0$: the coefficients are real symmetric and Lipschitz, and the Dirichlet realization has no zero eigenvalue by Poincaré's inequality. Positivity and contractivity follow from their Proposition 5.1(b).

Lemma 3.2. *For every $\lambda \in \mathbb{C} \setminus (-\infty, 0]$, $\lambda + \mathcal{N}_c$ is invertible on $C(\Gamma)$. Its resolvent is compact. For each $\varepsilon > 0$, $\lambda(\lambda + \mathcal{N}_c)^{-1}$ is uniformly bounded when $0 < |\lambda| \le R$ and $|\arg \lambda| \le \pi - \varepsilon$, for any fixed finite R.*

Proof. For $a > 0$, contractivity and the semigroup identity give

$$R_a := (a + \mathcal{N}_c)^{-1} = \int_0^\infty e^{-at} T(t)\, \mathrm{d}t. \tag{15}$$

For example, the difference quotient of the right-hand side $Q_a f$ is

$$\frac{T(h)Q_a f - Q_a f}{h} = \frac{e^{ah} - 1}{h} Q_a f - \frac{e^{ah}}{h} \int_0^h e^{-as} T(s) f \, \mathrm{d}s \longrightarrow a Q_a f - f.$$

It gives $Q_a f \in D(\mathcal{N}_c)$ and $(a + \mathcal{N}_c) Q_a f = f$. Injectivity follows from the energy identity (2).

Every $T(t)$, $t > 0$, is compact by its continuous kernel and Ascoli. Strong continuity is uniform on the compact image of $T(t/2)$, so T is norm continuous for positive times. Truncating (15) to $[\delta, M]$ and using norm Riemann sums proves that R_1 is compact. For $\lambda \notin (-\infty, 0]$, $I + (\lambda - 1)R_1$ is injective: a kernel vector belongs to $D(\mathcal{N}_c)$ and satisfies $(\lambda + \mathcal{N}_c)\varphi = 0$, whereas

$$\lambda \|\varphi\|_2^2 + \int_\Omega C \nabla u \cdot \nabla \overline{u} = 0 \tag{16}$$

forces $\varphi = 0$ at such a parameter. The compact Fredholm alternative therefore gives

$$(\lambda + \mathcal{N}_c)^{-1} = R_1 [I + (\lambda - 1)R_1]^{-1}. \tag{17}$$

One elementary form of that alternative is as follows. An injective $I - K$ with compact K is bounded below, by a compactness argument on a sequence of unit vectors. Its iterated ranges are closed. If its range were proper, these ranges would strictly decrease; unit vectors separated from the next range would make their images under K uniformly separated, contradicting compactness.

Let $P\varphi = \sigma(\Gamma)^{-1} \int_\Gamma \varphi \, \mathrm{d}\sigma$, regarded as a constant function. Connectedness of Ω and (16) give $\ker \mathcal{N}_c = \mathbb{C}1$. The self-adjoint L^2 realization fixes constants and preserves the boundary integral; hence R_1 commutes with P. On $X_0 = \ker P$, the compact Fredholm alternative makes $I - R_1$ invertible. Since $\mathcal{N}_c R_1 = I - R_1$, the restriction of $\mathcal{N}_c$ to X_0 has bounded inverse $R_1(I - R_1)^{-1}$. The Neumann series at zero therefore bounds $\lambda(\lambda + \mathcal{N}_c)^{-1}$ on X_0 for small $|\lambda|$; on constants it equals P. On the remaining compact parameter set, the resolvent identity and (17) give norm continuity and a finite bound. □

3.3 The large-parameter estimate

Proposition 3.3. *For every $\varepsilon \in (0,\pi)$ there is a finite M_ε such that*

$$\|\lambda(\lambda+\mathcal{N}_c)^{-1}\|_{C(\Gamma)\to C(\Gamma)} \le M_\varepsilon \qquad (\lambda \neq 0,\ |\arg\lambda| \le \pi-\varepsilon),$$

where $\mathcal{N}_c$ is the part of the weak Dirichlet-to-Neumann operator in $C(\Gamma)$.

Proof. By Lemma 3.2, a failure would give parameters $|\lambda_j| \to \infty$ with $|\arg\lambda_j| \le \pi-\varepsilon$, data $f_j \in C(\Gamma)$, and $\varphi_j = (\lambda_j+\mathcal{N}_c)^{-1} f_j$ such that

$$\frac{\|f_j\|_\infty}{|\lambda_j|\|\varphi_j\|_\infty} \longrightarrow 0. \tag{18}$$

Choose $q_j \in \Gamma$ attaining the supremum and set $\tau_j = \overline{\varphi_j(q_j)}/|\varphi_j(q_j)|$. Let u_j be the harmonic extension of $\tau_j\varphi_j/\|\varphi_j\|_\infty$, so that $u_j(q_j)=1$. The weak maximum principle, applied to real phase components, gives $\|u_j\|_\infty \le 1$. Its proof on this bounded domain is the usual test with the positive part above the maximum boundary value, which has zero trace.

Pass to one Lipschitz chart containing the q_j with a fixed margin, and rotate coordinates once so that the domain is above its graph. Set

$$r_j = |\lambda_j|^{-1}, \qquad v_j(y) = u_j(q_j + r_j y), \qquad z_j = r_j\lambda_j.$$

On every fixed bounded set, for large j the scaled domain is $D_j = \{y_d > \psi_j(y')\}$, where $\psi_j(0)=0$ and the Lipschitz constants are bounded by a common L. A diagonal subsequence converges locally uniformly to a global Lipschitz function ψ. The coefficients $C_j(y) = C(q_j + r_j y)$, in the fixed rotated coordinates, converge uniformly locally on D_j to a constant positive matrix C_*. Indeed, q_j has a convergent subsequence and the Lipschitz modulus of C at scale r_j tends to zero.

The scaled weak equation is

$$\int_{D_j} C_j\nabla v_j\cdot\nabla\overline{\eta} + z_j\int_{\partial D_j} v_j\overline{\eta}\,\mathrm{d}\sigma_j = \int_{\partial D_j} g_j\overline{\eta}\,\mathrm{d}\sigma_j, \tag{19}$$

for compactly supported tests in the chart, where

$$g_j(y) = \frac{r_j\tau_j f_j(q_j+r_j y)}{\|\varphi_j\|_\infty} \quad (y\in\partial D_j).$$

Thus $\|g_j\|_\infty \to 0$ by (18). The factor r_j in the boundary terms comes from dividing the weak equation by r_j^{d-2}, while surface measure scales by r_j^{d-1}. After a subsequence $z_j \to z_*$, where $|z_*|=1$ and $|\arg z_*| \le \pi-\varepsilon$. The conormal derivatives $g_j - z_j v_j$ are uniformly bounded. Lemma 3.1 gives reflected ambient extensions with uniform local H^1 and Hölder bounds. Weak H^1 compactness and Ascoli on successive compact sets yield a bounded limit v on $D=\{y_d>\psi(y')\}$, locally H^1 up to its boundary, with $v(0)=1$.

In graph coordinates set $a_j(x) = \sqrt{1+|\nabla\psi_j(x)|^2}$. A further diagonal weak-$*$ subsequence gives

$$a_j \overset{*}{\rightharpoonup} b \quad \text{locally in } L^\infty, \qquad 1 \le b \le \sqrt{1+L^2}. \tag{20}$$

Uniform convergence of the extensions and of ψ_j gives uniform convergence of $v_j(x,\psi_j(x))$ on compact sets. These values can therefore multiply the weak-$*$ convergence in (20). In the bulk, $1_{D_j}\to 1_D$ almost everywhere and strongly in every finite local L^p space. Together with coefficient convergence and weak convergence of the extension gradients, this passes the bulk integral in (19). The result is

$$\int_D C_*\nabla v\cdot\nabla\overline{\eta} + z_*\int_{\mathbb{R}^{d-1}} b(x)v(x,\psi(x))\overline{\eta(x,\psi(x))}\,\mathrm{d}x = 0. \tag{21}$$

Initially this holds for compact smooth ambient tests. They are dense in the local graph H^1 test space: reflect across the graph, multiply by an ambient cutoff, mollify the extension, and restrict back to D. Trace continuity passes the boundary term as well.

Flatten the limiting graph by $\Phi(x,t) = (x, t + \psi(x))$. Its determinant is one, so the matrix in the half-space equation is

$$A(x) = (D\Phi)^{-1} C_*(D\Phi)^{-T}.$$

It is real symmetric uniformly elliptic and independent of t. Equation (21) becomes precisely (13) with $U = v \circ \Phi$, $\lambda = z_*$ and the weight b in (20). Proposition 2.4 gives $U = 0$, contradicting $v(0) = 1$. This proves the large-parameter estimate and hence (4). □

4 Holomorphic extension

We finish with the contour argument, including the behavior at the vertex of the sector. Put $X = C(\Gamma)$. With $\mu = -\lambda$, Proposition 3.3 says that, for each $\omega > 0$,

$$R(\mu) = (\mu - \mathcal{N}_c)^{-1}, \qquad \|R(\mu)\|_{X\to X} \le \frac{M_\omega}{|\mu|} \quad (\mu \neq 0,\ |\arg \mu| \ge \omega). \tag{22}$$

Proof of Theorem 1.1. Fix $0 < \omega < \pi/2$ and $r > 0$. Let $G_{\omega,r}$ run inward from infinity on the ray $\arg \mu = \omega$, follow the circle of radius r counterclockwise from angle ω to $2\pi - \omega$ through the negative half-plane, and then run outward on the ray $\arg \mu = -\omega$. For $|\arg z| < \pi/2 - \omega$, define

$$E(z) = \frac{1}{2\pi i} \int_{G_{\omega,r}} e^{-z\mu} R(\mu)\, \mathrm{d}\mu. \tag{23}$$

The integral converges in operator norm. Cauchy's theorem in the annular region outside the positive sector makes it independent of r. Banach-valued integration and Cauchy's theorem can be applied in norm here; see [8, Theorems 3.27, 3.29 and 3.31].

Fix $\theta < \pi/2 - \omega$ and estimate with $r = 1/|z|$. On either ray $\mathrm{Re}(z\mu) \ge c_\theta |z||\mu|$. Each ray contribution is at most $M_\omega \int_1^\infty e^{-c_\theta s}\, \mathrm{d}s/s$. On the circle, $|e^{-z\mu}| \le e$ and its length is $(2\pi - 2\omega)r$. Thus E is bounded for all $z \in \Sigma_\theta$. Keeping r fixed locally in z, differentiation under the integral proves operator norm holomorphy.

For real $a > r$, absolute convergence permits integrating $e^{-at}E(t)$ in t first. We obtain

$$\int_0^\infty e^{-at} E(t)\, \mathrm{d}t = \frac{1}{2\pi i} \int_{G_{\omega,r}} (a + \mu)^{-1} R(\mu)\, \mathrm{d}\mu = (a + \mathcal{N}_c)^{-1}.$$

For the last equality close the contour on the left. Its orientation is clockwise, and the pole at $\mu = -a$ has residue $R(-a) = -(a + \mathcal{N}_c)^{-1}$; the outer arc tends to zero by (22). This agrees with the Laplace transform of $T(t)$ in (15), so uniqueness gives $E(t) = T(t)$ for $t > 0$. For completeness, scalar uniqueness for bounded continuous functions follows by the substitution $s = e^{-t}$: vanishing Laplace transforms give all sufficiently high integer moments on $(0, 1)$. Polynomial density in $C[0, 1]$ then gives zero against continuous tests and hence zero everywhere by continuity on $(0, 1)$. Bounded linear functionals separate the Banach-valued functions.

The identity theorem extends the semigroup law from positive real parameters to $E(z + w) = E(z)E(w)$ on each such sector, first in one variable and then in the other. The sector is convex and closed under addition. Constructions with different ω agree on overlaps, and their union is $\Sigma_{\pi/2}$. Denote the resulting extension by $T(z)$.

Finally, for $\varphi \in D(\mathcal{N}_c)$ the sector bound makes $\int_0^z T(w)\mathcal{N}_c\varphi\, \mathrm{d}w$ a holomorphic primitive from the vertex. One may integrate first from an interior point and use Cauchy's theorem;

small connecting arcs at zero have norm $O(\text{radius})$, so the limit is path independent. On positive real parameters the semigroup derivative gives this primitive as $\varphi - T(z)\varphi$. The identity theorem gives the same equality throughout the sector, and hence

$$\|T(z)\varphi - \varphi\|_\infty \le M_\theta |z| \|\mathcal{N}_c \varphi\|_\infty \qquad (z \in \Sigma_\theta).$$

Density of $D(\mathcal{N}_c)$ in X, together with (3), proves strong continuity at zero for every $\varphi \in X$. □

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